Letter for Number

Year 4 Pt 2 · **Lesson Worksheet**

Worked Examples

Example 1: Using A=1, B=2, C=3 … Z=26, what is the value of F + K?

Answer: 17 — F = 6 and K = 11, so 6 + 11 = 17.

Example 2: Using A=1, B=2, … Z=26, decode the word given by the numbers 4-15-7.

Answer: DOG — 4 = D, 15 = O, 7 = G.

Example 3: If A=5, B=6, C=7 … (each letter's value is 4 more than its usual alphabet position), what is the value of the letter D?

Answer: 8 — D is normally the 4th letter, and 4 + 4 = 8.

Purple — Accessible (15 questions)

1. Using A=1, B=2, C=3 … Z=26, what is the value of the letter J?
2. Using A=1, B=2, … Z=26, what letter has the value 14?
3. Using A=1, B=2, … Z=26, what is the value of the letter M? A) 11 B) 12 C) 13 D) 14 E) 15
4. Using A=1, B=2, … Z=26, what is B + E?
5. Using A=1, B=2, … Z=26, what is H − C?
6. Using A=1, B=2, … Z=26, decode the word given by the numbers 3-1-20. A) BAT B) CAT C) CAR D) BAR E) CAN
7. Using A=1, B=2, … Z=26, encode the word DOG as a sequence of numbers.
8. Using A=1, B=2, … Z=26, what is the value of the letter R?
9. Using A=1, B=2, … Z=26, select ALL the letters below with a value GREATER than 20. A) S B) U C) W D) T E) V
10. Using A=1, B=2, … Z=26, decode the word given by the numbers 19-21-14.
11. Using A=1, B=2, … Z=26, what is F + G? A) 11 B) 12 C) 13 D) 14 E) 15
12. Using A=1, B=2, … Z=26, what is the value of the letter Y?
13. Using A=1, B=2, … Z=26, encode the word CAT as a sequence of numbers.
14. Using A=1, B=2, … Z=26, what is P − J? A) 4 B) 5 C) 6 D) 7 E) 8
15. Using A=1, B=2, … Z=26, decode the word given by the numbers 2-9-7.

Green — Standard (15 questions)

1. Using A=1, B=2, … Z=26, what is C + H + M?
2. Using A=1, B=2, … Z=26, what is the value of Z − A?
3. Using A=1, B=2, … Z=26, decode the word given by the numbers 16-12-1-14. A) PLAT B) PLAN C) PLEN D) PLON E) PLIN

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4. Using A=1, B=2, ... Z=26, what is the SUM of the values of the letters in the word CAB?

5. Using A=1, B=2, ... Z=26, encode the word STAR as a sequence of numbers.

6. Using A=1, B=2, ... Z=26, what is K × 2? A) 20 B) 22 C) 24 D) 26 E) 28

7. Using A=1, B=2, ... Z=26, what is the value of the letter that comes immediately after N in the alphabet?

8. Using A=1, B=2, ... Z=26, what is the DIFFERENCE between the values of W and Q?

9. Using A=1, B=2, ... Z=26, select ALL the words below whose letters have a total SUM greater than 30. A) CAT (3,1,20) B) DOG (4,15,7) C) BAT (2,1,20) D) SUN (19,21,14) E) PIG (16,9,7)

10. Using A=1, B=2, ... Z=26, what is the value of the MIDDLE letter of the word FOX?

11. Using A=1, B=2, ... Z=26, what is the value of V + V? A) 40 B) 42 C) 44 D) 46 E) 48

12. Using A=1, B=2, ... Z=26, decode the word given by the numbers 7-18-5-5-14.

13. Using A=1, B=2, ... Z=26, what is the value of the letter that comes 3 places AFTER J in the alphabet?

14. Using A=1, B=2, ... Z=26, what is the SUM of the values of the letters in the word TEN? A) 37 B) 38 C) 39 D) 40 E) 41

15. Using A=1, B=2, ... Z=26, encode the word MOON as a sequence of numbers.

Red — Challenge (15 questions)

1. If A=5, B=6, C=7, ... (each letter's value is 4 more than its usual alphabet position), what is the value of the letter D?

2. Using the same rule as above (A=5, B=6, C=7, ...), what is the value of J?

3. If A=3, B=4, C=5, ... (each letter's value is 2 more than its usual alphabet position), what is the value of the letter H? A) 8 B) 9 C) 10 D) 11 E) 12

4. If A=0, B=1, C=2, ... (each letter's value is 1 less than its usual alphabet position), what is the value of the letter M?

5. Using the rule A=0, B=1, C=2, ... (1 less than usual position), what letter has the value 19?

6. If Z=1, Y=2, X=3, ... (the alphabet is reversed, so A=26), what is the value of the letter C? A) 22 B) 23 C) 24 D) 25 E) 26

7. Using the reversed rule (Z=1, Y=2, ..., A=26), what letter has the value 2?

8. If A=5, B=6, C=7, ... (4 more than usual position), what is the SUM of the values of D and E under this rule?

9. If A=14, B=15, C=16, ... (each letter's value is 13 more than its usual alphabet position), select ALL the letters below with a value of 20 or MORE under this rule. A) F B) K C) J D) C E) H

10. Using the rule A=14, B=15, C=16 ... (13 more than usual position), what is the value of the letter Z?

11. If A=2, B=4, C=6, ... (each letter's value is DOUBLE its usual alphabet position), what is the value of the letter G? A) 12 B) 14 C) 16 D) 18 E) 20

12. Using the doubling rule (A=2, B=4, C=6, ... value = 2 × usual position), what letter has the value 20?

13. Using A=1, B=2, ... Z=26 (the STANDARD rule), decode the word given by the numbers 23-9-14. Then, using the DOUBLING rule from the previous question (value = 2 × usual position), give the three numbers that would represent the SAME word.

14. Using A=1, B=2, ... Z=26, the word GOLD is encoded as 7-15-12-4. If each of these four numbers is increased by 1, what four letters do the NEW numbers represent? A) HPME B) HPMF C) HPLE D) GPME E) HOME

15. Using A=1, B=2, ... Z=26, find a single letter X such that the value of X plus the value of C equals the value of M.

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